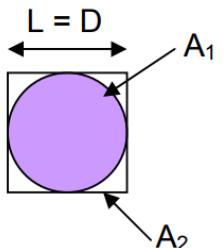
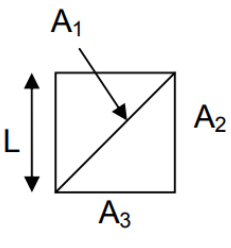
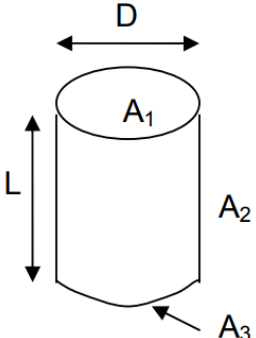


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE End Semester Examination (Regular) – December 2019 Branch: B.Tech in Mechanical Engineering Subject Name with Code: Heat Transfer [BTMEC 501] Max Marks: 60 Sem: V Date: 09/12/2019 Duration: 3 hr.		
Instructions to the Students: 1. Solve ANY FIVE questions out of the following six questions. 2. Each question carries 12 marks. 3. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of each question. 4. Use of non-programmable scientific calculator is allowed. 5. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)
Q. 1	Solve the following:	Marks
A)	Define overall heat transfer coefficient. How is it related to thermal resistance? Write down equation for total thermal resistance for composite hollow cylinder made up of two layers (with thermal conductivity k_1, k_2 with thickness b_1, b_2) and convective boundary conditions on either sides of composite cylinder with convective heat transfer coefficients h_1, h_2 . Assume some notation for radii like r_1, r_2, r_3 .	(CO-2) 6
B)	A composite wall of 1m^2 surface area is constructed of two layers. The first layer is of 1 cm thick steel ($k = 45 \text{ W/m K}$) and the second layer of 10 cm thick fiberglass insulation ($k = 0.035 \text{ W/m K}$). Determine: a) thermal resistance of the composite wall b) overall heat transfer coefficient.	(CO-1) 6
Q.2	Solve the following:	
A)	Derive equation for critical radius of insulation for a cylinder. A 2 mm diameter electric wire at 46°C is covered by 0.5 mm thick plastic insulation ($k = 0.03 \text{ W/m K}$). The insulation of the wire is exposed a medium at 10°C with convective heat transfer coefficient of $20.0 \text{ W/m}^2 \text{ K}$. Determine the critical insulation thickness. Will plastic insulation dissipate max heat?	(CO-2) 6
B)	Write equations for Biot number & Fourier number. Aluminium sphere weighing 5.5 kg and initially at a temperature of 290°C is suddenly immersed in a fluid at 15°C . The convective heat transfer coefficient is $58 \text{ W/m}^2\text{K}$. Estimate the time required to cool the aluminium to 95°C . Use lumped capacity method for calculation. Assume following property values for aluminium: $\rho = 2700 \text{ kg/m}^3$ $k = 205 \text{ W/m K}$ $C_p = 900 \text{ J/kg K}$	(CO-2) 6
Q. 3	Solve the following:	
A)	Sketch laminar and turbulent boundary layers (BL) for flow over a flat plate. Also, show velocity profiles within the BL in the two regions: a) in the laminar region b) in the	(CO-4) 6

	turbulent region. Assume uniform velocity profile on the upstream side of the plate. State the value of Re at transition.		
B)	Air at 60°C and atmospheric pressure flows over a thin flat plate (<i>one side</i>) which is 1 m wide and 2 m in length. The free stream air velocity is 1 m/s. Calculate a) thickness of velocity boundary layer at a distance $x = 1.5$ m and b) total drag force on the plate. Use following values of fluid properties: $\nu = 18.96 \times 10^{-6} \text{ m}^2/\text{s}$, $\rho = 1.06 \text{ kg/m}^3$ Use following equations for thickness of velocity boundary layer and drag coefficient, respectively: $\frac{\delta}{x} = \frac{5}{\sqrt{Re_x}} \quad \text{and} \quad C_f = 1.328 Re_L^{-0.5}$	(CO-4)	6
Q.4	Solve the following:		
A)	Water at 50°C enters 1.5 cm diameter tube of a heat exchanger. Assume velocity of water at mean temperature as 1 m/s. The tube surface is maintained at 90°C. Calculate the exit water temperature if the length of tube is 2 m. Assume following properties of water at mean temperature (<i>neglect variation in properties with temperature</i>): $\mu = 489.2 \times 10^{-6} \text{ kg/(m.s)}$ $\rho = 984.4 \text{ kg/m}^3$ $k = 0.656 \text{ W/m K}$ $c_p = 4178 \text{ J/kg K}$ Use following correlations: $f = 0.079 Re^{-0.25}$ $Nu_D = \frac{(f/2)[Re_D - 1000].Pr}{1 + 12.7 \left[\frac{f}{2} \right]^{0.5} (Pr^{2/3} - 1)}$	(CO-4)	6
B)	A 15 cm diameter horizontal iron pipe with 1 m length is exposed to saturated steam at 100°C on inside and still air at 20°C on outside. Calculate the convective heat transfer rate from the outer surface of pipe and compare it with radiant heat transfer rate if the surrounding surfaces (imaginary) are at 20°C. Assume outer surface of the pipe as black. Use following correlation for horizontal pipe: $Nu = 0.48 Ra^{0.25}$ Use following values of properties of air at 60°C: $\rho = 1.06 \text{ kg/m}^3 \quad \mu = 20.1 \times 10^{-6} \text{ N.s/m}^2, \quad k = 0.029 \text{ W/m K} \quad C_p = 1005 \text{ J/kg K}$	(CO-4)	6
Q. 5	Solve the following:		
A)	In a double-pipe counter-flow heat exchanger 10,000 kg/h of an oil having a specific heat 2095 J/kg.K is cooled from 80°C to 50°C by 8000 kg/h of cooling water at 25°C. Assume overall heat transfer coefficient of 300 W/m²K and specific heat of water as 4180 J/kg.K. a) Determine LMTD b) Determine the heat exchanger area.	(CO-5)	6
B)	Draw the pool boiling curve & identify different boiling regimes. Define critical flux & show corresponding point on pool boiling curve.		6

Q. 6	Solve the following:		
A)	Consider a 20 cm diameter spherical ball at 800 K suspended in air. Assuming the ball closely approximates a blackbody, determine: a) The total blackbody emissive power b) The amount of radiant energy emitted by the ball in 5 min. c) The monochromatic blackbody emissive power at a wavelength of 3 micrometers.	(CO-6)	6
B)	State various shape factor relations (algebra) in radiation heat transfer.	(CO-6)	6
	*** End ***		

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch :Mechanical Engineering Semester : V Subject Code & Name: BTMC 501 & Heat Transfer Max Marks: 60 Date: 28/01/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Air at 20 °C blows over a 50 cm x 75 cm hot plate at 250 °C. The film heat transfer coefficient is 25 W/m ² .K. 300 W is lost from the plate surface by radiation. Calculate heat transfer rate and other side plate temperature. Thermal conductivity of the plate material is 43 W/m .K. The plate is 2 cm thick.	(L3/CO1)	6
B)	What is thermal conductivity? List the factors affecting the thermal conductivity.	(L1/CO2)	6
C)	An insulating powder is densely packed in the annular space between two concentric spheres with radii 75 mm and 50 mm. The inner sphere is uniformly heated with electric power input of 30 W. Steady state temp attained by the inner sphere is 120 °C and that by outer surface is 30 °C. Neglecting the thermal resistance offered by the spheres: a) Draw analogous electrical cct diagram b) Calculate thermal conductivity of the powder	(L3/CO1)	6
Q.2	Solve Any Two of the following.		12
A)	Derive an equation to find heat dissipation from an infinitely long-fin.	(L2/CO3)	6
B)	Explain in brief, initial and boundary conditions.	(L1/CO2)	6
C)	Derive general heat conduction equation for Cartesian coordinate system.	(L1/CO1)	6
Q. 3	Solve Any Two of the following.		12
A)	Write a note on a) Forced Convection b) Free Convection c) Radiation.	(L1/CO1)	6
B)	A hot rectangular plate 5 cm X 3 cm maintained at 200 °C is exposed to still air at 30 °C. Calculate percentage increase in convective heat transfer rate if smaller side of the plate is held vertical than the bigger side. Neglect ITG of the thickness. Use Correlation $Nu=0.59 (Gr.Pr)^{0.25}$	(L3/CO4)	6

	Air properties at 115 °C: density = 0.91 kg/m ³ ; C _p =1.009 kJ/kg K; $\mu=22.65 \times 10^{-6}$ N s/m ² ; k=0.0331 W/m K.		
C)	Water flows at 360 kg/hr. through a metallic tube of 10 mm diameter and 3 m length. It enters the tube at 25 °C. Outer surface of the tube is maintained at a constant temp of 100 °C. Calculate the exit temp of the water. Properties of water: $\mu=5.62 \times 10^{-4}$ kg/m s; C _p =4174 J/kg K; k=0.664 W/m K. Use the following correlation: $N_u=0.023 Re^{0.8} Pr^{0.4}$ for turbulent flow $N_u=3.66$ for laminar flow	(L3/CO4)	6
Q.4	Solve Any Two of the following.		12
A)	Derive the expression for LMTD method for an analysis of parallel flow heat exchanger.	(L2/CO5)	6
B)	Write a short note on Overall Heat Transfer Coefficient for plate heat exchanger.	(L1/CO5)	6
C)	A double pipe parallel flow heat exchanger use oil (C _p = 1.88 kJ/kg.K) at an initial temperature of 205°C to heat water, flowing at 225 kg/hr. from 16°C to 44°C. The oil flow rate is 270 kg/hr. a) What is the heat transfer area required for an overall heat transfer coefficient of 340 W/m ² .K. b) Determine the number of transfer unit (NTU). c) Calculate the effectiveness of the heat exchanger.	(L3/CO5)	6
Q. 5	Solve Any Two of the following.		12
A)	Determine the view factors F ₁₂ and F ₂₁ for the following geometries:  1  2  3 1) Sphere of diameter D inside a cubical box of length L = D. 2) Diagonal partition within a long square duct. 3) End and side of a circular	(L3/CO6)	6

	tube of equal length and diameter, $L = D$.		
B)	State various shape factor relations (algebra) in radiation heat transfer.	(L1/CO6)	6
C)	Find out heat transfer rate due to radiation between two infinitely long parallel planes. One plane has emissivity of 0.4 and is maintained at 200 °C. Other plane has emissivity of 0.2 and is maintained at 30 °C. If a radiation shield ($\epsilon=0.5$) is introduced between the two planes, find percentage reduction in heat transfer rate and steady state temp of the shield.	(L2/CO6)	6
	*** End ***		

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Course: B. Tech.

Branch : Mechanical Engineering

Semester : V

Subject Code & Name: BTMC 501 & Heat Transfer

Max Marks: 60

Date: 07/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

- A) Write a note on thermal conductivity. Give some examples of heat transfer in engineering. (L1/CO1) 6
- B) The inner surface of furnace wall is at 200 °C and outer surface is at 50 °C. Calculate the heat loss per m² area of the wall. If the thermal conductivity of the brick is 0.5 W/m °C and wall thickness is 200 mm. (L3/CO1) 6
- C) What is the law of heat conduction? State the assumptions on which this law is based? (L1/CO1) 6

Q.2 Solve Any Two of the following.

12

- A) Derive general heat conduction equation in Cartesian co-ordinate system. (L2/CO2) 6
- B) Define Nusselt number, Prandtl number and Grashoff number. (L2/CO2) 6
- C) Write a short note on critical radius of insulation. (L1/CO2) 6

Q. 3 Solve Any Two of the following.

12

- A) Consider the flow of a gas with density 1 kg/m³, viscosity 1.5×10^{-5} kg/(m.s), specific heat $C_p = 846$ J/(kg.K) and thermal conductivity $k = 0.01665$ W/(m.K), in a pipe of diameter $D = 0.01$ m and length $L = 1$ m, and assume the viscosity does not change with temperature. The Nusselt number for a pipe with (L/D) ratio greater than 10 and Reynolds number greater than 20000 is given by “ $Nu = 0.026 Re^{0.8} Pr^{1/3}$ ”. While the Nusselt number for a laminar flow for Reynolds number less than 2100 and $(Re Pr D/L) < 10$ is “ $Nu = 1.86 [Re Pr (D/L)]^{1/3}$ ”.
If the gas flows through the pipe with an average velocity of 0.1 m/s, the heat transfer coefficient is? (L1/CO3) 6
- B) What is convective heat transfer? Distinguish between free and forced Convection (L2/CO4) 6

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|----|--|----------|---|
| C) | Explain the concept of hydrodynamic and thermal boundary layers with reference to flow over a flat heated plate. | (L3/CO5) | 6 |
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Q.4	Solve Any Two of the following.		12
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|----|--|----------|---|
| A) | What is heat exchanger and types of heat exchanger (any 05)? | (L2/CO5) | 6 |
| B) | Derive the expression for LMTD method for an analysis of counter flow heat exchanger. | (L3/CO2) | 6 |
| C) | Water enters a parallel flow double-pipe heat exchanger at 15 °C, flowing at the rate of 1200 kg/hr. It is heated by oil ($C_p = 2000 \text{ J/kg.K}$), flowing at the rate of 500 kg/hr from an inlet temperature of 90 °C. For an area of 1 m ² and an overall heat transfer coefficient of 1,200 W/m ² .K determine the total heat transfer and the outlet temperatures of water and oil. | (L3/CO3) | 6 |

Q. 5	Solve Any Two of the following.		12
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|----|---|----------|---|
| A) | Define shape factor. Explain relations/theorems of shape factor. | (L2/CO5) | 6 |
| B) | What is Stefan Boltzmann Law? Explain the concept of total emissive power of a surface. | (L1/CO6) | 6 |
| C) | Two very large parallel plates with emissivity 0.5 exchange heat. Determine the percentage reduction in the heat transfer rate if a polished aluminium radiation shield of emissivity = 0.25 is placed in between the plates. | (L3/CO6) | 6 |

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Regular & Supplementary Winter Examination-2023

Course: B.Tech.

Branch: Mechanical Engineering/Mech(Sandwich)

Subject Name: Heat Transfer (Code: BTMC501)

Semester: V

Max Marks: 60

Date of Exam: 01 Jan 2024

Duration: 3 hrs.

	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Attempt Any Two of the following.		12
A)	Derive an equation for heat transfer rate through a hollow cylinder by applying Fourier's Law of heat conduction. The hollow cylinder has: • length L, • inner radius r_i, outer radius r_o • inner surface temperature T_i and outer surface temperature T_o, thermal conductivity k. $T_i > T_o$	(L4/CO1)	6
B)	An insulating powder is densely packed in the annular space between two concentric spheres with radii 75 mm and 50 mm. The inner sphere is uniformly heated with electric power input of 30 W. Steady state temperature attained by inner sphere is 120°C and that by outer sphere is 30°C. <i>Neglect thermal resistance offered by the material of the spheres.</i> a) Calculate thermal conductivity of the powder. b) Calculate the thermal resistance offered by insulating powder.	(L3/CO1)	6
C)	Explain the mechanism of heat conduction. Explain "why is a good conductor of electricity also a good conductor of heat?"	(L4/CO1)	6
Q.2	Attempt Any Two of the following.		12
A)	A solid steel ball 5 cm in diameter and initially at 450°C is quenched in a controlled environment at 90°C with convective heat transfer coefficient of 115 W/m ² K. Determine the time taken by center of the ball to reach a temperature of 150°C. Use following properties of steel ball $c_p = 420 \text{ J/kg.K}$, $\rho = 8000 \text{ kg/m}^3$, $k = 46 \text{ W/m.K}$. Justify the equation used for temperature distribution.	(L3/CO2)	6
B)	State general heat conduction equation in Cartesian coordinate system and write down the situations/conditions for its use.	(L2/CO2)	6

C)	Calculate fin efficiency and heat loss from a fin (<i>assuming insulated fin-tip</i>) with circular cross-section under following conditions: Air temperature : 27°C Temperature of fin at the base : 227°C Thermal conductivity of a fin : 220 W/m K h for a cylindrical surface of a fin : 15 W/m ² K Fin diameter : 1 cm Fin length : 3 cm	(L4/CO3)	6
Q. 3 Attempt Any Two of the following.			12
A)	A hot rectangular plate 5 cm x 3 cm maintained at 200°C is exposed to still air at 30°C. Calculate convective heat transfer rate by natural convection from both sides of the plate if the smaller side of the plate is held vertical. Use following correlation: $Nu = 0.59 Ra^{0.25}$ Use following Air properties at mean film temperature of 115°C Density = 0.91 kg/m ³ , $c_p = 1.009$ kJ/kg K, $\mu = 22.65 \times 10^{-6}$ N.s/m ² , $k = 0.0331$ W/m K	(L4/CO4)	6
B)	Sketch laminar and turbulent boundary layers (BL) for flow over a flat plate.	(L4/CO4)	6
C)	Natural convection heat transfer is governed by Nusselt number, Grashoff number and Prandtl number. State equations for these dimensionless numbers.	(L3/CO4)	6
Q.4 Attempt Any Two of the following.			12
A)	Derive expression for mean temperature difference (LMTD) for parallel flow heat exchanger.	(L4/CO5)	6
B)	Sketch the temperature variations along the length of the heat exchangers with following flow arrangements: 1. Parallel flow 2. Counter flow Write down equation for Overall Heat Transfer Coefficient for a tube-in-tube heat exchanger considering two convective resistances on two sides of a single-layer inner-tube and considering conductive resistance offered by the tube material.	(L3/CO5)	6
C)	In a double-pipe counter-flow heat exchanger 10,000 kg/h of an oil having a specific heat 2095 J/kg.K is cooled from 80°C to 50°C by 8,000 kg/h of cooling water at 25°C. Assume overall heat transfer coefficient of 300 W/m ² K and specific heat of water as 4180 J/kg.K. a) Determine LMTD b) Determine area of the heat exchanger.	(L4/CO5)	6
Q. 5 Attempt Any Two of the following.			12
A)	State any four shape factor relations (algebra) used in radiation heat transfer.	(L2/CO6)	6

B)	State the any two laws of radiation heat transfer.	(L2/CO6)	6
C)	Two large parallel plates, held at temperature 727°C and 227°C, have emissivity values of 0.9 and 0.6 respectively. • Calculate the heat transfer rate by radiation per square meter of plate area (<i>without radiation shield</i>). • Calculate the heat transfer rate by radiation per square meter of plate area if a radiation shield having emissivity 0.4 is placed between the two plates. • Calculate the % reduction in heat transfer rate by use of a radiation shield.	(L3/CO6)	6
	*** End ***		

BL – Bloom’s Taxonomy Levels (L1- Remembering, L2- Understanding, L3 – Applying, L4 – Analyzing, L5 – Synthesizing, L6 - Evaluating)
CO – Course Outcomes